

THE IMPACT OF SOCIOECONOMIC INDICATORS ON GDP PER CAPITA ACROSS COUNTRIES

by
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DECLARATION

I, B.M.Sachini Chamodya , declare that the presented project report titled, "THE IMPACT OF SOCIOECONOMIC INDICATORS ON GDP PER CAPITA ACROSS COUNTRIES" is uniquely prepared by me based on the group project carried out under the supervision of Dr. K.D.Prasangika Department of Mathematics, Faculty of Science, University of Ruhuna, as a partial fulfillment of the requirements of the level II Case Study I course unit (MIS2231) of the Bachelor of Science Financial Mathematics and Industrial Statitics Degree in Department of Mathematics, Faculty of Science, University of Ruhuna, Sri Lanka. It has not been submitted to any other institution or study program by me for any other purpose.

Signature:.....

Date:.....

SUPERVISOR'S RECOMMENDATION

I/We certify that this study was carried out by B.M.Sachini Chamodya under my/our supervision.

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ABSTRACT

Using multiple linear regression analysis, this study examines the socioeconomic factors that influence the GDP in 154 different countries. The objective of the study is to examine how the socioeconomic factors (Child mortality rate, Exports, Health, Imports, Income, Inflation rate, Life expectancy, and Total fertility rate) affect the GDP per capita of different countries. Initial data exploration includes summary statistics, box plots, and correlation matrices to assess variable distributions and relationships. From the results that were obtained it can be concluded that income, life expectancy, and exports were positively associated with GDP per capita, while child mortality and fertility rate showed strong negative associations. The findings of this study will highlight the importance of socioeconomic indicators in shaping national economic performance.

Keywords: GDP per capita, child mortality, income, inflation, health, import, export, multiple linear regression

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CHAPTER 1
Introduction

1.1 Background of the Study

Gross Domestic Product per capita (GDP per capita) is one of the most commonly used indicators to measure the nation's economic health. This study focuses on a selected set of socioeconomic indicators as independent variables to understand their relationship with GDP per capita. These variables are namely exports, imports, net income, child mortality, health expenditure, inflation, life expectancy, and total fertility rate. This research aims to identify which factors have the most effects on GDP per capita by modeling these relationships using multiple linear regression. By analyzing the relationships between these variables, the research will provide a deeper understanding about the most valuable information on specific factors that can affect on economic growth of a country .

1.2 Problem Statement

As we commonly know, one of the main factors that measures the economic growth of a country is the GDP per capita .This research seeks to explore how different socioeconomic factors work together to impact GDP per capita across 154 countries by analyzing that factors: Inflation, Income, Export, Health, Total fertility rate, Life expectancy, Child mortality Rate in 154 countries. Our goal is identified which are the factors that most mostly effect on the country's economic growth

1.3 Objective of the study

The main objective of this study is to identify the relationship between seriocomic indicators and the GDP per capita of different countries by seeing the weights of the independent variables.

1.3.1 Objectives of the Study

1. To identify the relationship between Child mortality rate and the GDP per capita.
2. To identify the relationship between Inflation rate and the GDP per capita.
3. To identify the relationship between Total fertility rate and the GDP per capita.
4. To identify the relationship between Income and the GDP per capita of a country.
5. To identify the relationship between Imports and the GDP per capita of a country.
6. To identify the relationship between Exports and the GDP per capita of a country.
7. To identify the relationship between Life expectancy and the GDP per capita.
8. To identify the relationship between Health and the GDP per capita of a country

1.4 Research Questions

1. How does Child mortality rate influence GDP per capita across different countries?
2. How does Inflation rate influence GDP per capita across different countries?
3. How does Total fertility rate influence GDP per capita across different countries?
4. How does income per capita influence GDP per capita across different countries?
5. How does Imports influence GDP per capita across different countries?
6. How does exports influence GDP per capita across different countries?
7. How does Life expectancy influence GDP per capita across different countries?
8. How does health influence GDP per capita across different countries?

1.5 Significance of the study

By using multiple linear regression analysis, the study targets to give an understanding on the relationship between the GDPP and its determinants in a quantifiable way. The study's significance lies in its ability to identify which factors have the most impact on GDP per capita. Thereby highlighting areas where targeted measures could lead to meaningful economic improvements. It also supports global development goals by helping identify the key socioeconomic indicators that improve sustainable economic growth.

1.6 Hypothesis

H_0 : Null hypothesis

There is no linear relationship between dependent variable and independent variables.

H_1 : Alternative hypothesis

There is a linear relationship between dependent variable and independent variables.

CHAPTER 2
Literature Review

GDP per person is a common way to see how much money the average person in a country earns and how good their life is. Many studies have looked at how GDP per person is connected to different social and money things.

For example, Sari and Cahyadin (2021) studied Indonesia from 1986 to 2018. They found that when people live longer, GDP per person goes up. They also found a surprising link between baby death rates and GDP in Indonesia, which was not expected. When it comes to health spending, the results were mixed: in developing countries, inefficiencies can reduce the benefits of health investments, while wealthier countries usually see better results from spending more on health. Education also works hand in hand with health to improve productivity (Schultz, 2005).

The researchers used a special model called ARDL-ECM to look at how things change in both the short term and the long term. They also used dynamic panel model method , to compare results across countries. Their study showed that health and life expectancy is very important for Indonesia's economy. Therefore they think the government should make plans and policies to improves child survival and life expectancy . This may help the country grow and become stronger.

Another study by Batrancea et al. (2021) looked at 34 African countries from 2001 to 2019, It was found that exports had the biggest impact on GDP per capita where 1% rise in exports led to a 0.1% increase in GDP per capita. Also Imports had a positive and smaller effect. Foreign direct investment (FDI) helped increase capital formation by 0.12% while domestic savings showed a modest positive relationship. These results were related with earlier findings by Gossel and Biekpe (2014), showed that exports has a strong relationship with GDP growth.When researchers used a panel data analysis method using EViews 11, they found that the best ways to predict economic growth in African countries are by looking at exports, capital formation, and savings. These factors seem to have the strongest effect on how much the economy grows. However, the effect of foreign direct investment (FDI) was not as clear or strong as the others.

In another study, Bhide and Khanolkar (2020) started with 17 different factors to see which ones affect GDP per person the most. By using multiple regression analysis, they were able to find the five most important factors. These were: Net migration, Infant mortality, Number of phones per 1,000 people ,Literacy rate, Death rate.These five factors turned out to be the best at explaining why some countries have higher GDP per person than others.

Research by Divya and Devi (2014) highlights that education and infrastructure for the example the number of phones per capita are key factors. Lehmann and Wohlrabe (2015) emphasized that regional employment and education levels are crucial for GDP. These studies show that education, healthcare, technology that all have a strong impact on GDP per capita.

Looking at Kazakhstan, Berdimurat et al. (2024) analyzed data from 2012 to 2022 using correlation and linear regression. They found that average wages, the Human Development Index (HDI), fertility rate, and real interest rate are important predictors. This matches what Elistia and Syahzuni (2018) found in ASEAN countries, where HDI was strongly linked to GDP per capita.

Research by Çekrezi (2022) and Štilić (2023) also shows that fertility rates and interest rates as key factors. Berdimurat's team concluded that average salary has the biggest positive effect, while fertility rate, real interest rate, and somewhat unexpectedly HDI showed significant negative links.

Many existing studies have examined the relationship between individual socioeconomic factors such as income, trade, education, health and fertility. Most of these analyses have been limited to specific countries, regions, or smaller sets of variables. This study aims to fill this gap by analyzing a wider range of socioeconomic variables across varied sample of nations to provide a more understanding of their influence on GDP per capita.

CHAPTER 3
Data

3.1 About the data set

The dataset is about a broad view into the socioeconomic determinants of GDP per capita across a sample of 154 countries. The dataset contains figures representing each country's economic, health and demographic characteristics.

3.2 Metadata

The source of these data is Kaggle.

(https://www.kaggle.com/datasets/samira1992/countries-intermediate-dataset)

The data collection took place at 9.50 am on 9th April 2025

Independent variables	**Dependent variables**
Child mortality rate	GDP per capita
Imports	
Exports	
Inflation rate	
Income	
Life expectancy	
Health	
Total fertility rate	

Table 3.1: List of Variables

Variable	**Variable Type**
GDP per capita	Numerical
Child mortality rate	Numerical
Imports	Numerical
Exports	Numerical
Inflation rate	Numerical
Income	Numerical
Life expectancy	Numerical
Health	Numerical
Total fertility rate	Numerical

Table 3.2: List of Variables and Their Types

3.3 Data Dictionary

1. **Gdpp**

   The GDP per capita (calculated as the total GDP divided by the total population)

2. **Export**

   Exports of goods and services per capita (presented as a percentage)

3. **Child Mortality Rate**

   Number of deaths of children under 5 years of age per 1000 live births.

4. **Health**

   Total health spending per capita (presented as a percentage of GDP per capita).

5. **Imports**

   Imports of goods and services per capita (given as a percentage of GDP per capita).

6. **Income**

   Net income per person.

7. **Inflation Rate**

   The annual growth rate of the total GDP.

8. **Life Expectancy**

   The average number of years a newborn child would live.

9. **Total Fertility Rate**

   The average number of children per woman.

3.4 Data set preparation

Check NA values

Then we check is this data set has NA values by using this R code `is.null(data)`. Here also found that this data set hasn't any NA values.

Check NULL values

Thenwe check whether this data set has null value by using this R code `is.na(data)`. We found that this data set hasn't any null values.

Transformation of variables

We applied data transformation to selected independent variables.The child mortality rate and total fertility rate were transformed by taking their inverse values, which helped linearize their relationship with the dependent variable.

Outliers

Additionally, during the data preprocessing stage, some extreme outliers were identified and removed to reduce their influence on the regression results.

CHAPTER 4
Methodology

4.1 Conceptual model

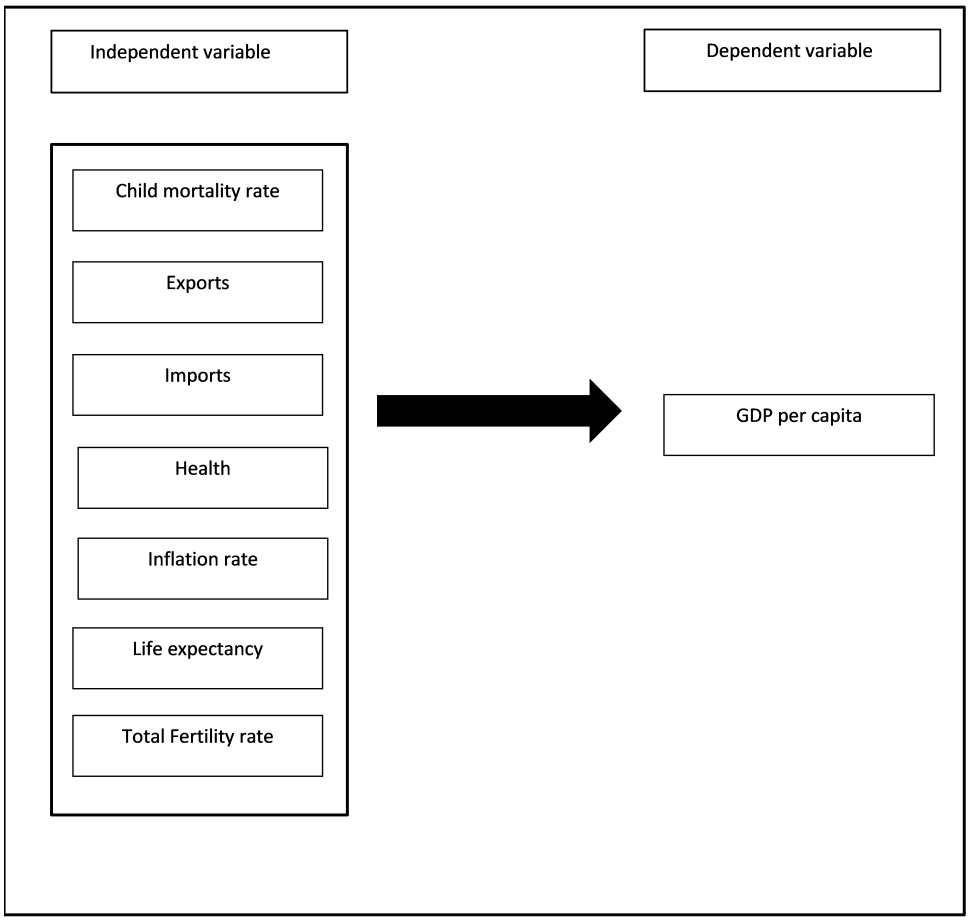


Figure 4.1: Conceptual Model

4.2 Research approach

The research approach for this study is quantitative research approach. The main target is to examine how socioeconomic factors affect the GDPP of countries. For this study multiple linear regression was used to investigate the relationship between the dependent variable and independent variables.

4.3 Research design

The primary goal of the study is to use multiple linear regression to assess how Child Mortality Rate , Exports , Health, Imports , Income , Inflation Rate, Life Expectancy, Total Fertility Rate impact on GDP per capita. This study has a quantitative research design to investigate the socioeconomic determinants of GDP per capita across a sample of 154 countries. The primary objective is to assess the relationship between the independent variables and the dependent variable.

The source of these data is Kaggle. Link to dataset

The data collection took place at 10.25 am on 5th June 2025.

Multiple linear regression model

Multiple linear regression is a statistical technique used to model the relationship between multiple independent variables and a dependent variable. A population model for a multiple linear regression model is written as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Where,

- Y is the dependent variable
- $X_1, X_2, \ldots, X_n$ are the independent variables
- β_0 is the intercept
- $\beta_1, \beta_2, \ldots, \beta_n$ are the regression coefficients

4.4 Significance of Regression

The test for significance of regression is a test to determine if there is a linear relationship between the response y and any of the regressor variables $x_1, x_2, \ldots, x_k$. The appropriate hypotheses are,

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_k = 0$$

$$H_1 : \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_k \neq 0$$

Rejection of this null hypothesis implies that at least one of the regressors $x_1, x_2, \ldots, x_k$ contributes significantly to the model.

Source of variation	Sum of squares	Degrees of freedom	Mean square	F_0
Regression	SS_R	k	MS_R	MS_R/MS_{res}
Residual	SS_{res}	$n-k-1$	MS_{res}	
Total	SS_T	$n-1$		

SS_R - Sum of squares of regression

SS_{res} - Sum of squares of residual

SS_T - Total sum of squares

The F_0 follows the $F_{k,n-k-1}$ distribution. Reject H_0 if

$$F_0 > F_{\alpha,k,n-k-1}.$$

4.5 Model Adequacy Assessment

In the fitting of the linear regression, a researcher will probably include independent variables that are not crucial for predicting the dependent variable. Therefore, we use the following methods to find the best model with the fewest variables that still explain a percentage of variance in the dependent variable that is comparable to the percentage explained with all the variables in the equation and compare the models. There are primarily three approaches for this process.

The forward Elimination

This approach begins with an empty equation and gradually adds the most significant predictor that is, the predictor variable with the lowest p-value until the model is not significantly improved by the addition of predictors.

The Backward Elimination

This method starts with the equation that includes the all predictor variables and step by step remove the least significant predictor that means the predictor variable with the highest p-value until the subtraction of predictors does not significantly change the model.

All Possible Regressions

Another term for this technique is the step wise selection method. This combines the backward elimination method and the forward selection method. Among these techniques, the backward elimination method is used in this study be cause it simplifies the model and supports the objective of analyzing the individual contributions of each variable to Sales growth of the online shipping company. Additionally, this method guarantees that all predictor variables that remain in the best regression equation are statistically significant, allowing for the interpretation of the results of the research.

4.5.1 Partial F test

A partial F-test is used to determine whether or not there is a statistically significant difference between a regression model and some nested version of the same model.

Hypothesis of the test,

$$H_0 : \text{Reduced model is suitable}$$

$$H_1 : \text{Full model is suitable}$$

A partial F-test calculates the following F test-statistic:

$$\text{Test statistic} \;=\; F_{\text{partial}} \;=\; \frac{SS_{\text{Reg(Full)}} - SS_{\text{Reg(Reduced)}}}{MS_{\text{Error(Full)}}}$$

And the test statistic compared with $F_{k+1,\,p,\,n-k-1}$.

4.5.2 Lack of fit and Pure error

- **Pure error**: Pure error occurs for repeated values of dependent variable.
- **Lack of fit**: Lack of fit happens when your model doesn't predict the outcome well. This can occur if you haven't included all the important factors in your model.

$$SS_E = SS_{\text{pure error}} + SS_{\text{Lack of fit}}$$

$$Test\ statistic = \frac{\text{Mean squares due to lack of fit}}{\text{Mean squares due to pure error}}$$

And the test statistic value is compared with the $F_{df.LF,df.PE}$.

If $F_{calculated} > F_{df.LF,df.PE}$, then the model is inadequate and needs an alternative model. If not, it is not necessary to find a more complicated model (Lack of fit is not significant).

Hypotheses of the test:

$$H_0 : \text{Model is adequate}$$

$$H_1 : \text{Model is inadequate}$$

CHAPTER 5
Results

5.1 Exploratory data Analysis

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3
gdpp	154	0	11334	1151	14289	231	1340	4670	13200
inv_total_fer	154	0	0.4328	0.0151	0.1879	0.1517	0.2602	0.4211	0.5626
inv_child_mort	154	0	0.08563	0.00717	0.08895	0.00840	0.01783	0.05222	0.11885
exports	154	0	36.03	1.58	19.58	0.11	22.50	30.45	49.82
life_exp	154	0	67.771	0.606	7.522	51.800	62.450	67.850	73.025
total_fer	154	0	2.876	0.116	1.443	1.030	1.777	2.375	3.842
inflation	154	0	6.686	0.499	6.187	-4.210	1.705	5.025	10.225
income	154	0	15189	1110	13772	609	3633	10180	22400
imports	154	0	45.98	1.52	18.89	11.80	30.87	43.30	58.90
health	154	0	6.906	0.198	2.463	1.970	5.070	6.710	8.815
child_mort	154	0	34.49	2.70	33.45	2.60	8.42	19.15	56.08

Variable	Maximum
gdpp	52100
inv_total_fer	0.9709
inv_child_mort	0.38462
exports	103.00
life_exp	82.000
total_fer	6.590
inflation	22.800
income	49400
imports	98.00
health	11.900
child_mort	119.00

Figure 5.1: Descriptive statistics Table

GDP per capita

The histogram and boxplot shows that the distribution appears to be strong right-skewed. The data does not follow a normal distribution. There are numerous outliers, and all of them are on the high end, indicating a long tail of exceptionally high values show that most countries in this dataset have a low GDP per capita.

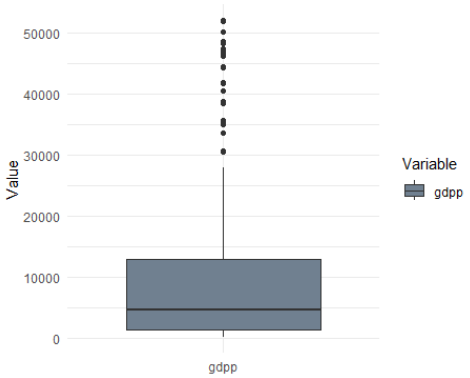


Figure 5.2: Boxplot of GDP per capita

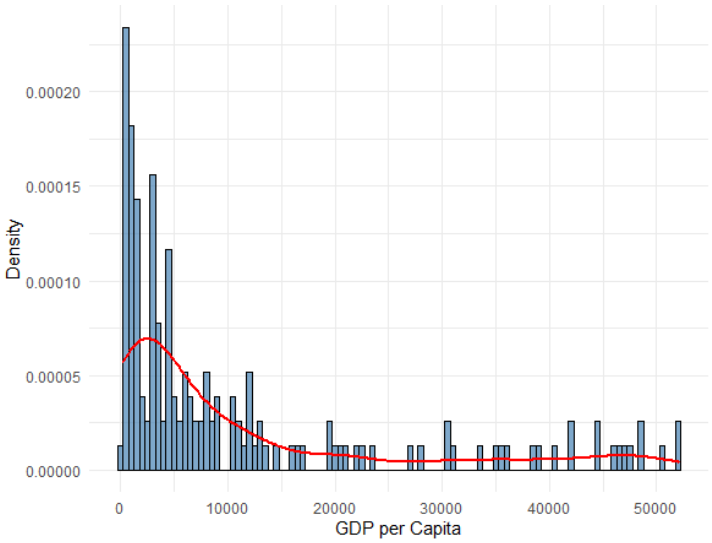


Figure 5.3: Histogram of GDP per capita

Export

Histogram show that the distribution of exports is not symmetrical and is moderately Right skewed and not follow a normal distribution. The regression line shows linear relationship and the scatter plot illustrates a strong linear relationship and positive correlation between exports and GDP per capita.

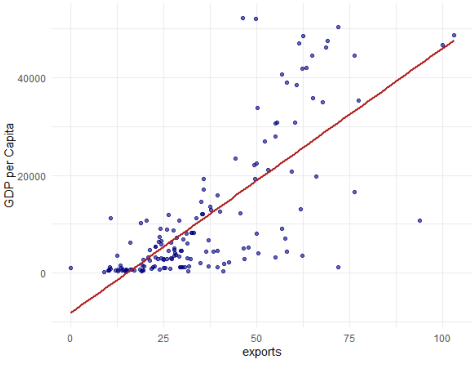


Figure 5.4: Scatter plot of Exports vs GDP per capita

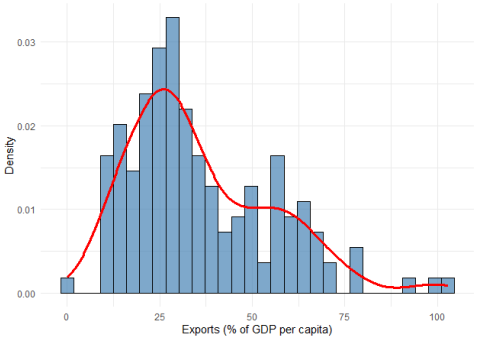


Figure 5.5: Histogram of Exports

Income

Based on the histogram ,the distribution is slightly negatively skewed / left-skewed and they clearly say that this distribution does not follow a normal distribution. The scatter plot clearly displays, there has a strong positive linear relationship with the two variablesand a positive correlation. The overall pattern strongly suggests that higher income levels are directly and linearly correlated with higher GDP per capita.

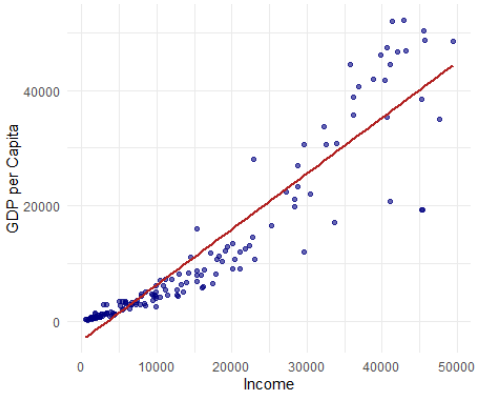


Figure 5.6: Scatter plot of Income vs GDP per capita

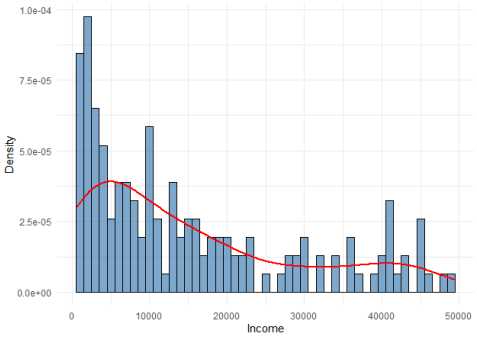


Figure 5.7: Histogram of Income

Imports

Histogram indicate that the distribution is slightly negatively skewed. Furthermore, the data does not follow a normal distribution. There is no strong linear relationship between "Imports" and "GDP per Capita" evident in this scatter plot. herefore, based on this visualization, imports do not appear to have a significant linear correlation with GDP per Capita.

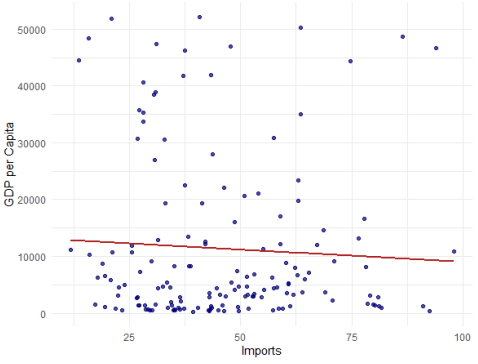


Figure 5.8: Scatter plot of Imports vs GDP per capita

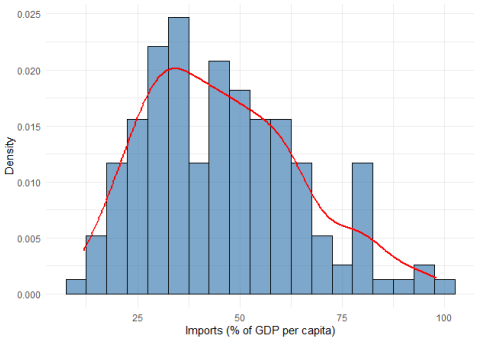


Figure 5.9: Histogram of Imports

Inflation

Based Histogram , the distribution is positively skewed. The data does not follow a normal distribution. There is a weak to moderate negative linear relationship between "Inflation" and "GDP per Capita" with a negative correlation in this scatter plot. suggesting that as inflation increases, GDP per Capita tends to decrease, though this relationship is not highly predictive.

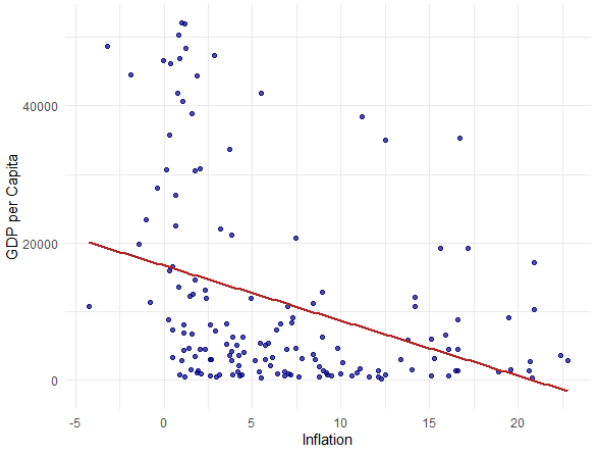


Figure 5.10: Scatter plot of Inflation vs GDP per capita

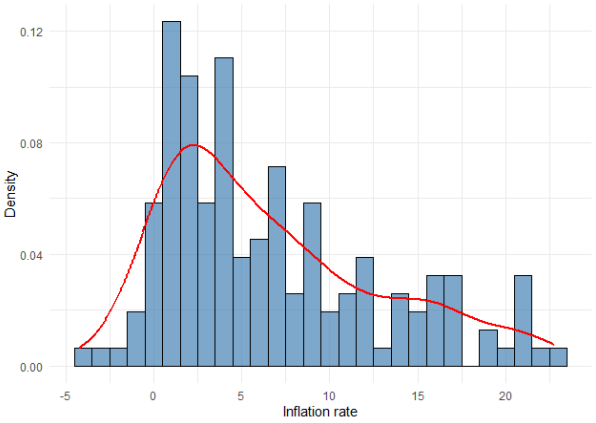


Figure 5.11: Histogram of Inflation

Health

The histogram confirms the distribution is approximately symmetric and does not show a strong skew and has no outliers. The scatter plot shows a positive relationship between health and GDP per capita.However the relationship is not perfect.

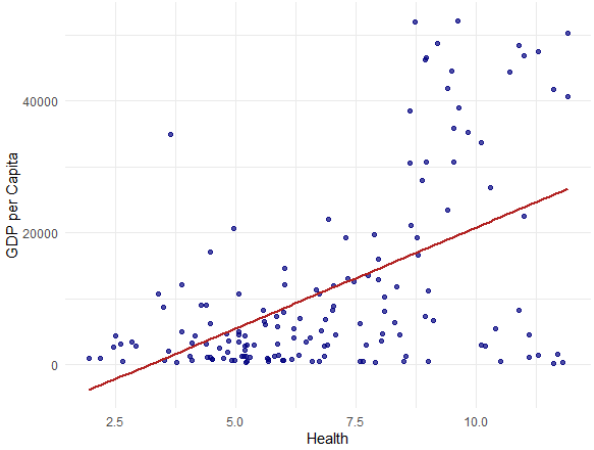


Figure 5.12: Scatter plot of Health vs GDP per capita

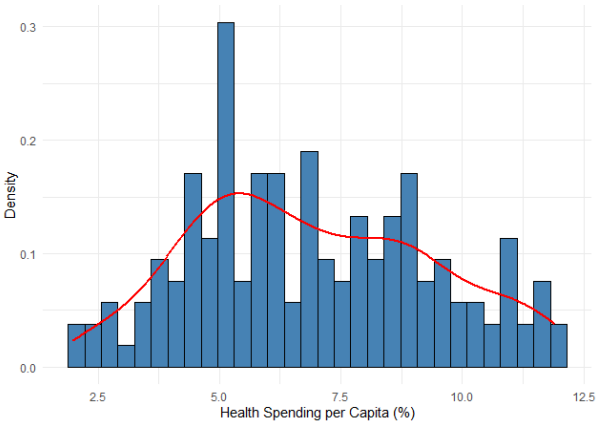


Figure 5.13: Histogram of Health

Child mortality

We used the inverse of child mortality to make its relationship with GDP per capita appear more linear on the scatter plot. Both histogram and boxplot indicate a strong positive skewness. This distribution does not follow a normal distribution. It clearly indicates a strong positive linear relationship with a strong positive correlation between the two variables. Therefore, based on this visualization, countries with fewer child deaths tend to have much higher economic output per person.

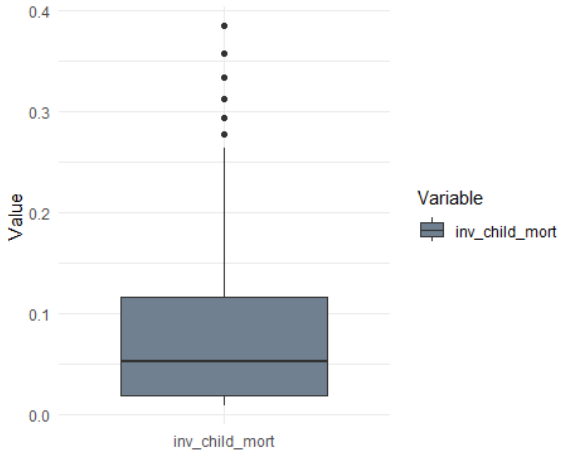


Figure 5.14: Boxplot of Inverse Child Mortality

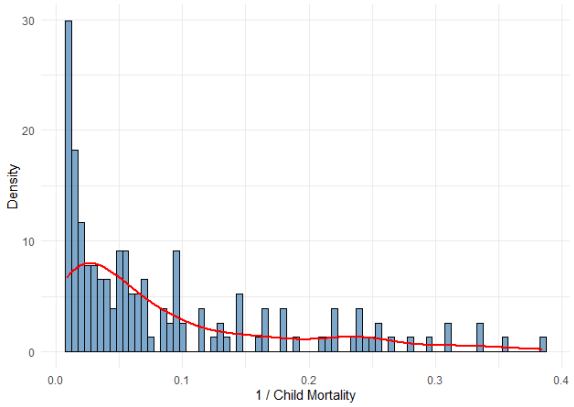


Figure 5.15: Histogram of Inverse Child Mortality

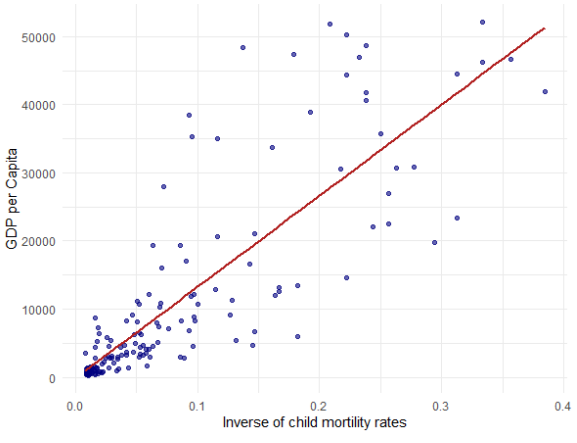


Figure 5.16: Scatter plot of Inverse Child Mortality vs GDP per capita

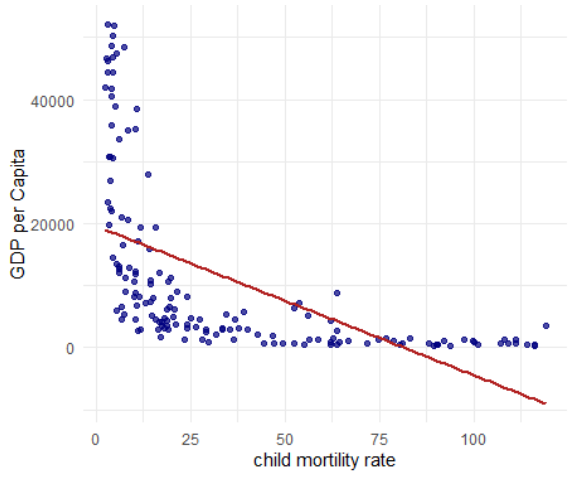


Figure 5.17: Scatter plot of Child Mortality vs GDP per capita

Total Fertility Rate

We used the inverse of total fertility rate to make its relationship with GDP per capita appear more linear . The histogram and boxplot show that the distribution is approximately symmetrical with positive (right) skewness.and the distribution does not follow a normal distribution. There is a positive linear relationship between the two variables. Therefore, based on this visualization, countries where people have fewer children (lower fertility rates) tend to have higher wealth per person.

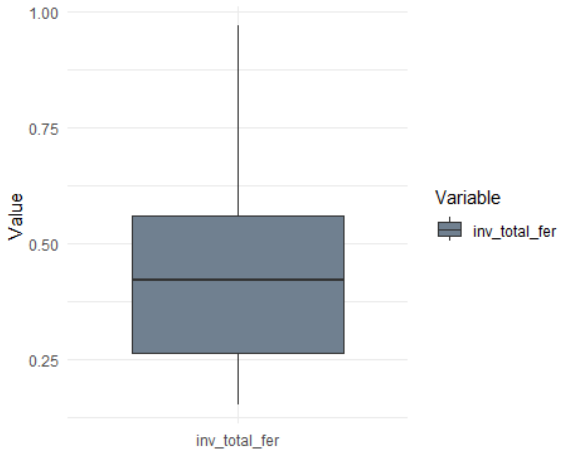


Figure 5.18: Boxplot of Inverse Total Fertility Rate

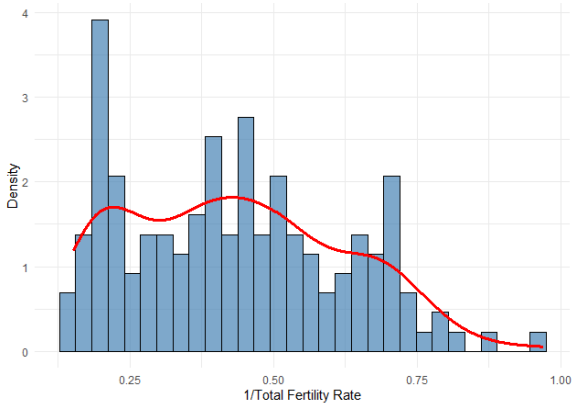


Figure 5.19: Histogram of Inverse Total Fertility Rate

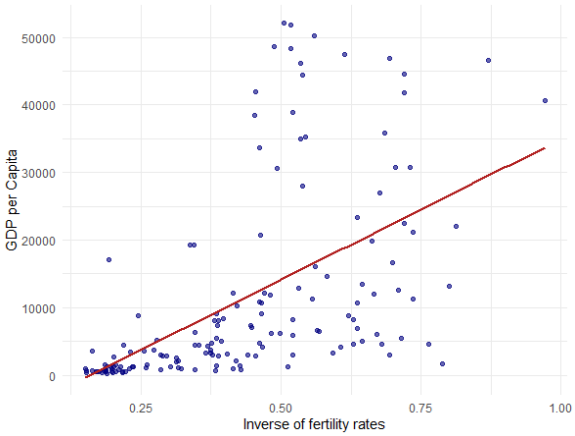


Figure 5.20: Scatter plot of Inverse Total Fertility Rate vs GDP per capita

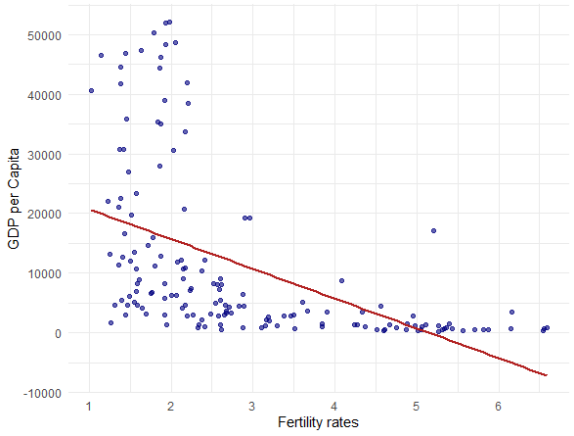


Figure 5.21: Scatter plot of Total Fertility Rate vs GDP per capita

Life expectancy

The boxplot is very slight negative skewed. Then the distribution does not strictly follow a normal distribution. It is showing a strong positive linear relationship between the two variables. The overall pattern strongly suggests that improvements in health, as reflected by life expectancy.

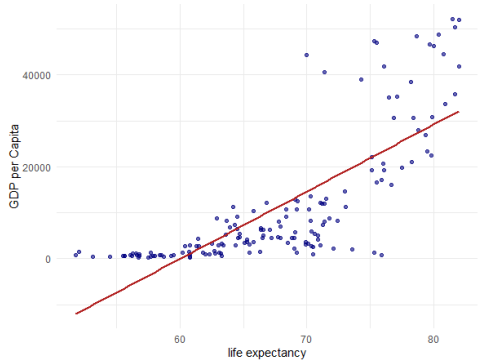


Figure 5.22: Scatter plot of Life Expectancy vs GDP per capita

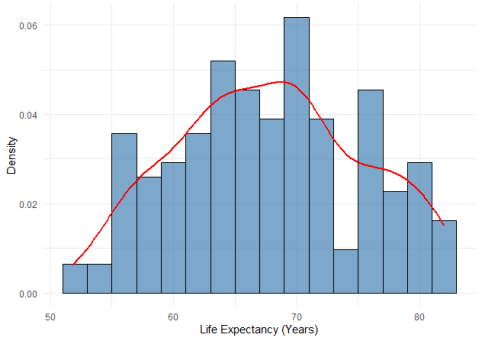


Figure 5.23: Histogram of Life Expectancy

5.1.1 Correlation Plot

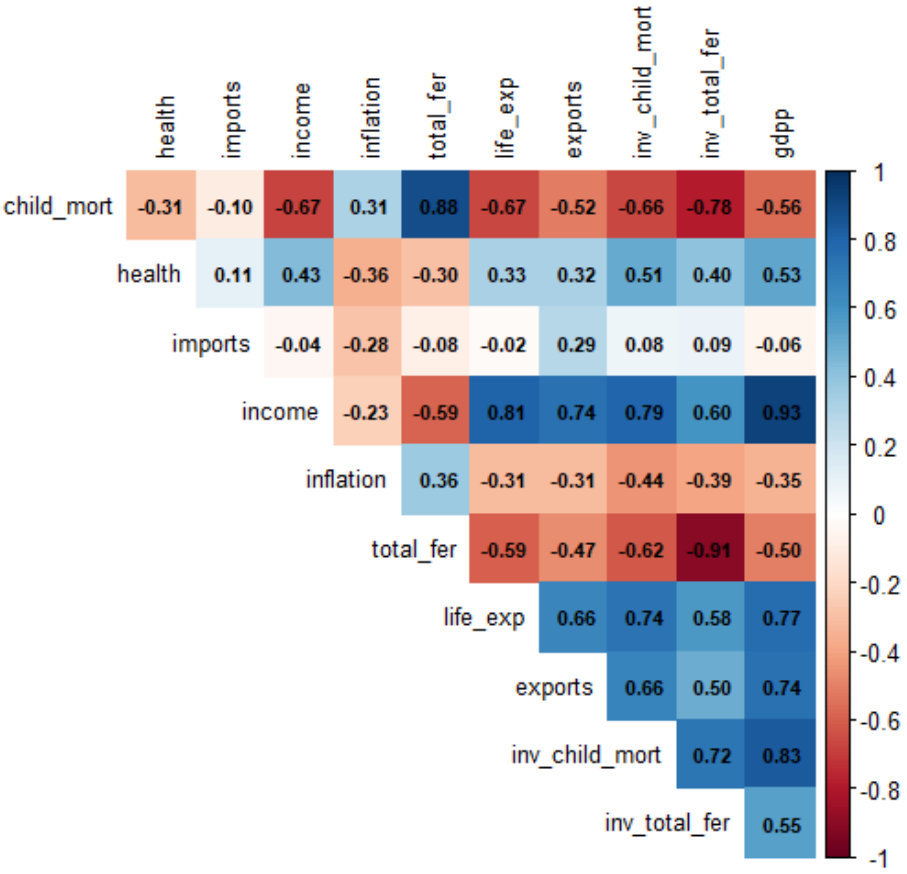


Figure 5.24: Correlation Plot of Variables

5.2 Quantitative Data Analysis

As mentioned earlier, there are no missing values and null values in our data set. In this quantitative data analysis part, we are trying to get the most suitable model for our data set and then we will get the idea about what factors are mainly effected to the sales growth of the online shopping company.

5.2.1 Correlation Analysis

	child_mort	health	imports	income	inflation	total_fer	life_exp	exports	inv_child	inv_fer	gdpp
child_mort	1.000	-0.312	-0.102	-0.670	0.312	0.880	-0.669	-0.518	-0.661	-0.785	-0.565
health	-0.312	1.000	0.106	0.434	-0.358	-0.298	0.326	0.324	0.505	0.402	0.528
imports	-0.102	0.106	1.000	-0.043	-0.284	-0.083	-0.024	0.290	0.078	0.085	-0.057
income	-0.670	0.434	-0.043	1.000	-0.234	-0.586	0.808	0.742	0.794	0.599	0.929
inflation	0.312	-0.358	-0.284	-0.234	1.000	0.365	-0.311	-0.311	-0.442	-0.392	-0.348
total_fer	0.880	-0.298	-0.083	-0.586	0.365	1.000	-0.590	-0.467	-0.618	-0.908	-0.504
life_exp	-0.669	0.326	-0.024	0.808	-0.311	-0.590	1.000	0.659	0.735	0.576	0.768
exports	-0.518	0.324	0.290	0.742	-0.311	-0.467	0.659	1.000	0.664	0.496	0.741
inv_child_mort	-0.661	0.505	0.078	0.794	-0.442	-0.618	0.735	0.664	1.000	0.723	0.833
inv_total_fer	-0.785	0.402	0.085	0.599	-0.392	-0.908	0.576	0.496	0.723	1.000	0.547
gdpp	-0.565	0.528	-0.057	0.929	-0.348	-0.504	0.768	0.741	0.833	0.547	1.000

Table 5.1: Correlation Matrix of Variables

According to this correlation matrix, exports, income, life expectancy, and the inverse of child mortality have a significant positive relationship with GDP per capita, indicating they are key contributors to economic performance. There is a moderately positive correlation between health spending and the inverse of the overall fertility rate, indicating some supportive influence. Also , child mortality and total fertility rate have a strong negative relationship with GDP per capita, reflecting that poorer health and higher fertility are linked to lower economic output. Meanwhile there are only weak or moderate relationships between health, imports, and inflation.

5.3 Regression Analysis

As mentioned in above our data set does not have any duplicate value and null values. Then we have tried to get a fitted model for the dependent variable that we have and our data set maybe have some outliers at the moment when we are doing for the fit.

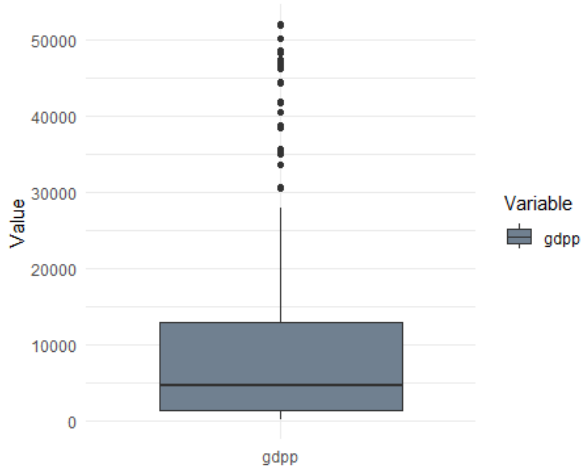


Figure 5.25: Boxplot of GDP per capita

We can not apply regression analyze if we can not satisfy it's assumptions. Those assumptions are

01. The residual values are normally distributed.

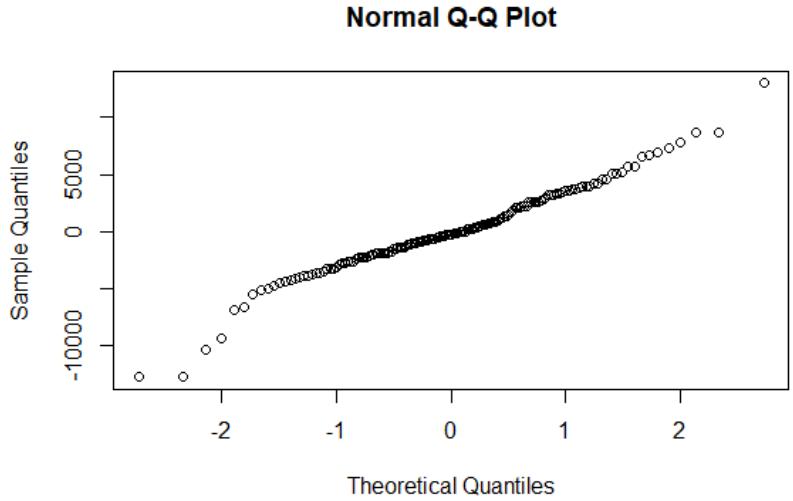


Figure 5.26: QQ plot of data

We can see this is normally distributed. We confirmed it by using "Shapiro.test". Then our this assumption is satisfied.

02. There must be a linear relationship between the dependent and the independent variables.

As we have shown scatter plots in Exploratory Data Analysis part we can see all independent variables and the dependent variable have a linear relationship. As well as in the correlation matrix (Table 5.1) also we can see the relationship between independent and dependent variables. Then our this assumption is also satisfied

03. Multicollinearity

If there are no collinearity in the data then all the values should be less than 0.8 . This assumption is verified by computing a correlation matrix among all the independent variables. In the (Table 5.1) we can see that the independent variables are not highly correlated with each other. Then our this assumption is also satisfied.

Variance Inflation Factor (VIF) value greater than 5 or 10 is generally considered indicative of problematic multicollinearity:

Variable	VIF
exports	3.016059
health	1.453825
imports	1.406635
income	5.358015
inflation	1.489495
life˙exp	3.290196
inverse child˙motility	4.377896
inverse total fer	2.154702

Table 5.2: VIF values for Predictor Variables

Most of the variables have VIF values well below the common threshold of 5.

However,income has a VIF of 5.36, slightly above the threshold of 5, suggesting a moderate level of multicollinearity.

No variable exceeds 10, so overall, the model does not exhibit severe multicollinearity.

04. Homoscedasticity

The homoscedasticity assumes that the variance of the residual error is similar across the value of each independent variable. We checked it by plotting predicted values against the standardized residual values.

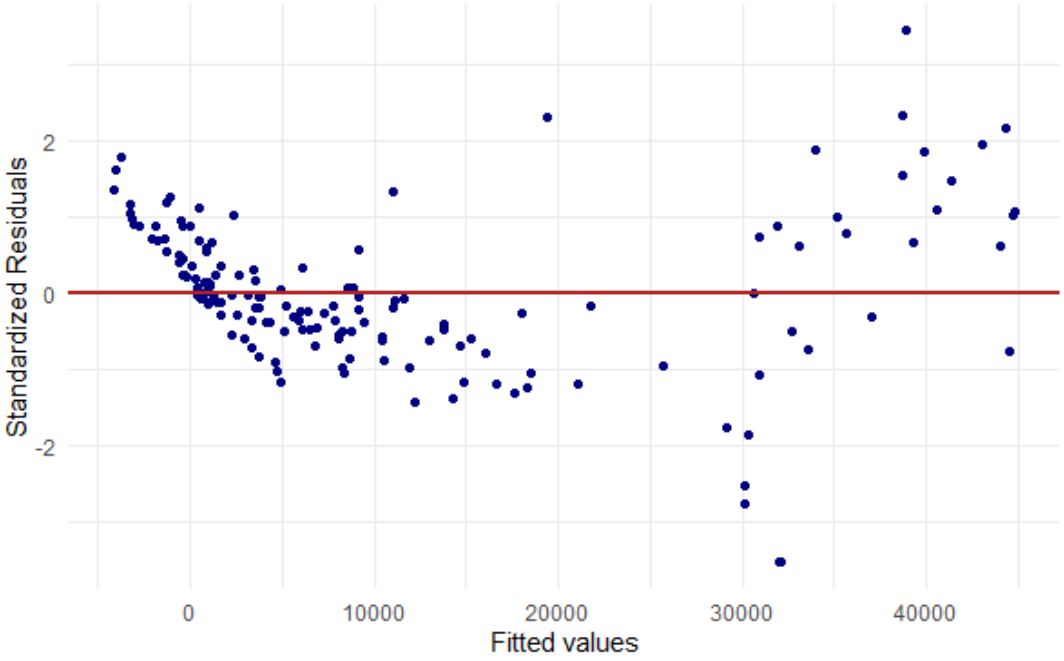


Figure 5.27: Homoscedasticity

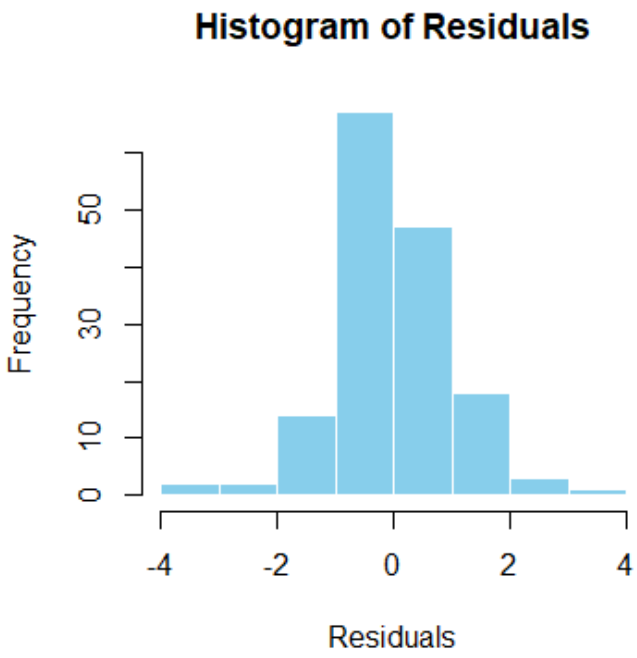


Figure 5.28: Histogram of Residuals

- The histogram appears to be roughly symmetric and bell-shaped.
- we can say residuals are approximately normally distributed, which is a good sign for the model's assumptions.

There are some outliers in this figure 5.25 , but considering our large observation we ignored them for our research. By satisfying these assumptions now we can apply multiple linear regression.

5.4 Estimate Model Parameters

In this section, relationship between the dependent and independent variables are mathematical represented. Then, which independent variables are to be included is chosen to assess the model's fit.

```
lm(formula = gdpp ~ exports + health + imports + income + inflation +
    life_exp + inv_child_mort + inv_total_fer, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-12968.3  -2287.7   -269.1   2443.6  13137.3

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     3.086e+03  5.178e+03   0.596 0.552178
exports         1.081e+02  2.840e+01   3.807 0.000207 ***
health          6.844e+02  1.567e+02   4.367 2.38e-05 ***
imports        -8.997e+01  2.010e+01  -4.475 1.53e-05 ***
income          6.770e-01  5.381e-02  12.581  < 2e-16 ***
inflation      -2.274e+02  6.316e+01  -3.600 0.000437 ***
life_exp       -4.464e+01  7.721e+01  -0.578 0.564061
inv_child_mort  4.213e+04  7.531e+03   5.594 1.07e-07 ***
inv_total_fer  -1.290e+04  2.501e+03  -5.156 8.13e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3960 on 145 degrees of freedom
Multiple R-squared:  0.9272,   Adjusted R-squared:  0.9232
F-statistic: 230.8 on 8 and 145 DF,  p-value: < 2.2e-16
```

Figure 5.29: Summary of full model

Our full model:

$$\begin{aligned}\text{GDP per capita} = {} & 3085.524 + 108.1128 \cdot \text{Exports} + 684.3999 \cdot \text{Health} \\ & - 89.96664 \cdot \text{Imports} + 0.6770 \cdot \text{Income} - 227.3570 \cdot \text{Inflation} \\ & - 44.63998 \cdot \text{Life Expectancy} + 42127.11 \cdot \text{Inverse child mortality} \\ & - 12897.03 \cdot \text{Inverse total fertility rate}\end{aligned}$$

```
Analysis of Variance Table

Response: gdpp
                Df     Sum Sq    Mean Sq   F value    Pr(>F)
exports          1 1.7166e+10 1.7166e+10 1094.4993 < 2.2e-16 ***
health           1 2.8948e+09 2.8948e+09  184.5685 < 2.2e-16 ***
imports          1 2.5917e+09 2.5917e+09  165.2438 < 2.2e-16 ***
income           1 5.3382e+09 5.3382e+09  340.3603 < 2.2e-16 ***
inflation        1 3.2401e+08 3.2401e+08   20.6585 1.148e-05 ***
life_exp         1 2.9419e+05 2.9419e+05    0.0188 0.8912550
inv_child_mort   1 2.3269e+08 2.3269e+08   14.8360 0.0001755 ***
inv_total_fer    1 4.1694e+08 4.1694e+08   26.5841 8.132e-07 ***
Residuals      145 2.2742e+09 1.5684e+07
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 5.30: ANOVA table of full model

Backward Elimination Method

Backward elimination method is used to fit the most suitable model for this data set.Backward elimination is a step wise variable selection method used in regression analysis to identify the most relevant independent variables that should be included in a regression model. It is a technique for simplifying the model while retaining its predictive power.Multiple linear regression usually uses backward elimination.The steps involved in the regression model's backward elimination procedure are shown below figure 5.30.

Stepwise Summary

Step	Variable	AIC	SBC	SBIC	R2	AdjR2
0	Base Model	3386.741	3392.815	2946.160	0.00000	0000
1	income	3081.831	3090.942	2642.693	0.86370	0281
2	inv_child_mort	3053.519	3065.667	2614.439	0.88806	0657
3	inv_total_fer	3039.047	3054.232	2600.056	0.89941	0740
4	health	3023.580	3041.801	2585.118	0.91020	0779
5	inflation	3016.803	3038.062	2578.692	0.91517	0231
6	imports	3009.942	3034.237	2572.423	0.91992	0665
7	exports	2997.608	3024.941	2561.522	0.92703	0353

Figure 5.31: Backward Elimination Model

After doing another two methods,that is All possible method and Backward method we also get the same result as forward method.

```
                          Model Summary
-----------------------------------------------------------------------
R                         0.963       RMSE                 3847.252
R-Squared                 0.927       MSE                14801348.963
Adj. R-Squared            0.924       Coef. Var                34.863
Pred R-Squared            0.915       AIC                    2997.608
MAE                    2911.685       SBC                    3024.941
-----------------------------------------------------------------------

                                  ANOVA
-----------------------------------------------------------------------------
                     Sum of
                    Squares         DF       Mean Square       F          Sg
-----------------------------------------------------------------------------
Regression    28959298103.576          7    4137042586.225    264.985    0.00
Residual       2279407740.326        146      15612381.783
Total         31238705843.903        153
-----------------------------------------------------------------------------
```

Figure 5.32: Reduced ANOVA Table

```
Model   => gdpp ~ exports + health + imports + income + inflation + life_exp+
           inv_child_mort + inv_total_fer
R2      => 0.927

Removed => life_exp

Model   => gdpp ~ exports + health + imports + income + inflation +
           inv_child_mort + inv_total_fer
R2      => 0.92703
```

Figure 5.33: Full Model VS Reduced model

So we can see that Life˙expectancy is not in our final regression Model.

Our reduced model :

$$\begin{aligned}\text{GDP per capita} = 260.296 &+ 106.139 \cdot \text{Exports} + 697.718 \cdot \text{Health} - 88.735 \cdot \text{Imports} \\ &+ 0.663 \cdot \text{Income} - 227.357 \cdot \text{Inflation} \\ &+ 41409.816 \cdot \text{Inverse Child Mortility} \\ &- 13000.211 \cdot \text{Inverse total fertility rate}\end{aligned}$$

Reduced Model Summary and ANOVA Table

```
Call:
lm(formula = gdpp ~ exports + health + imports + income + inflation +
    inv_child_mort + inv_total_fer, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-12928.2  -2223.3   -341.5   2377.6  13015.3

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     2.603e+02  1.708e+03   0.152 0.879113
exports         1.061e+02  2.813e+01   3.774 0.000233 ***
health          6.977e+02  1.547e+02   4.511 1.32e-05 ***
imports        -8.873e+01  1.995e+01  -4.449 1.70e-05 ***
income          6.635e-01  4.835e-02  13.722  < 2e-16 ***
inflation      -2.223e+02  6.240e+01  -3.562 0.000498 ***
inv_child_mort  4.141e+04  7.411e+03   5.587 1.10e-07 ***
inv_total_fer  -1.300e+04  2.489e+03  -5.222 5.97e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3951 on 146 degrees of freedom
Multiple R-squared:  0.927,    Adjusted R-squared:  0.9235
F-statistic:   265 on 7 and 146 DF,  p-value: < 2.2e-16
```

Figure 5.34: Reduced model summary

```
Analysis of Variance Table

Response: gdpp
                Df     Sum Sq    Mean Sq  F value    Pr(>F)
exports          1 1.7166e+10 1.7166e+10 1099.513 < 2.2e-16 ***
health           1 2.8948e+09 2.8948e+09  185.414 < 2.2e-16 ***
imports          1 2.5917e+09 2.5917e+09  166.001 < 2.2e-16 ***
income           1 5.3382e+09 5.3382e+09  341.919 < 2.2e-16 ***
inflation        1 3.2401e+08 3.2401e+08   20.753 1.095e-05 ***
inv_child_mort   1 2.1887e+08 2.1887e+08   14.019 0.0002597 ***
inv_total_fer    1 4.2581e+08 4.2581e+08   27.274 5.974e-07 ***
Residuals      146 2.2794e+09 1.5612e+07
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 5.35: Reduced Model ANOVA table

5.5 Model fit

Using R programming language we can get following result,

Model	Beta	Estimate	Std. Error	$Pr(>\|t\|)$
$(Intercept)$	$\hat{\beta}_0$	260.296	1708.425	0.879113
$exports$	$\hat{\beta}_1$	106.139	28.125	5.97e-07
$health$	$\hat{\beta}_2$	697.718	154.666	$<$ 2e-16
$imports$	$\hat{\beta}_3$	-88.735	19.946	1.10e-07
$income$	$\hat{\beta}_4$	0.663	0.048	0.000233
$inflation$	$\hat{\beta}_5$	-222.276	62.403	0.000498
inv_child_mort	$\hat{\beta}_6$	41409.816	7411.345	1.32e-05$\times 10^{-5}$
inv_total_fer	$\hat{\beta}_7$	-13000.211	2489.308	1.70e-05
$life_expt$	$\hat{\beta}_8$	0	20.9879	0.56406

5.6 Hypothesis Testing

Hypothesis 1

- **Null hypothesis (H0):**

$$\beta(i) = 0, \quad i = 1, 2, 3, 4, 5, 6, 7 \tag{5.1}$$

  There is no linear relationship between dependent variable and independent variables.

- **Alternative hypothesis (Ha):**

$$\beta(i) \neq 0, \quad i = 1, 2, 3, 4, 5, 6, 7 \tag{5.2}$$

  There is a linear relationship between dependent variable and independent variables.

Research question 1

Relationship between Exports and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for Inverse child mortility.Then we can reject the null hypothesis.

$$\beta_1 \neq 0,\ \beta_1 = 106.139 \tag{5.3}$$

Research question 2

Relationship between Health and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can reject the null hypothesis.

$$\beta_2 \neq 0,\ \beta_2 = 697.718 \tag{5.4}$$

Research question 3

Relationship between Imports and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can reject the null hypothesis.

$$\beta_3 \neq 0,\ \beta_3 = \text{-}88.735 \tag{5.4}$$

Research question 4

Relationship between Income and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can reject the null hypothesis.

$$\beta_4 \neq 0,\ \beta_4 = 0.663 \tag{5.4}$$

Research question 5

Relationship between Inflation and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can reject the null hypothesis.

$$\beta_5 \neq 0,\ \beta_5 = \text{-}227.357 \tag{5.4}$$

Research question 6

Relationship between Inverse child motility and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can reject the null hypothesis.

$$\beta_6 \neq 0,\ \beta_6 = 41409.816 \tag{5.4}$$

Research question 7

Relationship between Inverse total fertility rate and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can reject the null hypothesis.

$$\beta_7 \neq 0,\ \beta_7 = \text{-}13000.211 \tag{5.4}$$

Research question 8

Relationship between Life expectancy and the GDP per capita

According to the summary of full model we can see the P value is very small than 0.5 for discounts.Then we can not reject the null hypothesis.

$$\beta_8 \neq 0,\ \beta_8 = 0 \tag{5.4}$$

Hypothesis 2

- **Null hypothesis (H0):**

  Reduced model is suitable

- **Alternative hypothesis (Ha):**

  Full model is suitable

```
> anova(reduced_model,full_model)
Analysis of Variance Table

Model 1: gdpp ~ exports + health + imports + income + inflation +
         inv_child_mort + inv_total_fer

Model 2: gdpp ~ exports + health + imports + income + inflation + life_exp +
         inv_child_mort + inv_total_fer

     Res.Df    RSS        Df  Sum of Sq  F        Pr(>F)
1    146     2279407740
2    145     2274165314  1   5242426    0.3343   0.5641
```

Figure 5.36: F˙Partial

First, we should consider first the F partial.

$$F_{(\text{partial})} = \frac{SS_{\text{Reg (FULL)}}\text{-}SS_{\text{Reg (REDUCED)}}}{MS_{\text{Error (FULL)}}} \tag{5.10}$$

$$= \frac{[2279407740\text{-}2274165314]}{15683899}$$

$$= 0.3343$$

Then we get the F table value. It for 95 percent confidence interval is,

$$F(1\text{-}0.05, 1, 145) = 3.906392$$

The F partial $<$ F table, i.e. $0.3343 < 3.906392$

Then we do not reject null hypothesis at 0.05 significance level. Therefore,

The reduced model is suitable.

5.7 Interpretation of the reduced model

$$\begin{aligned}\text{GDP per capita} = 260.296 &+ 106.139 \cdot \text{Exports} + 697.718 \cdot \text{Health} - 88.735 \cdot \text{Imports} \\ &+ 0.663 \cdot \text{Income} - 227.357 \cdot \text{Inflation} \\ &+ 41409.816 \cdot \text{Inverse Child Mortility} \\ &- 13000.211 \cdot \text{Inverse total fertility rate}\end{aligned}$$

- **Intercept**: when all the predictors are equal to zero, the GDP per capita is increased by 260.296

- **Exports (106.139)**: A one-unit increase in Exports, the GDP per capita is predicted to be 106.139, assuming all other factors in constant.

- **Health (697.718):** A one-unit increase in health spending is predicted to increase GDP per capita by 697.718,assuming all other variables to be constant.

- **Imports (-88.735):** A one-unit increase in imports is associated with a decrease of 88.735 in GDP per capita, assuming all other variables to be constant.

- **Income (0.663):** For each one-unit increase in income, GDP per capita increases by 0.663,assuming all other factors being constant.

- **Inflation (-227.357):** A one-unit increase in inflation leads to a decrease of 227.357 in GDP per capita,assuming all other factors being constant.

- **Inverse Child Mortality (41409.816):** A one-unit increase in inverse child mortality is associated with a large positive increase of 41409.816 in GDP per capita,assuming all other factors being constant.

- **Inverse Total Fertility Rate (-13000.211):** A one-unit increase in inverse total fertility rate is associated with a decrease of 13000.211 in GDP per capita, assuming other all other factors being constant.

CHAPTER 6

6.1 Discussion

Our study aimed to predict, what socioeconomic factors that influence the GDP in 154 different countries. These variables are namely exports, imports, net income, child mortality, health expenditure, inflation, life expectancy, and total fertility rate. We began descriptive analysis, visualizing relationships through plots and summary statistics and checked it was not a normally distributed. Additionally, during the data preprocessing stage, some extreme outliers were identified and removed to reduce their influence on the regression results. Then plotted the histogram.

After that, we used data transformation for our independent variable, child mortality rate and total fertility rate were transformed by taking their inverse values, which helped linearize their relationship with the dependent variable.Then we analyzed our data set by using R software. We did correlation analysis, estimate models, model summaries, ANOVA tables by using this software.

When we considering our data set, we could see our R squared value of full model. It was 92.70 percent. To get best fitted model and R squared value, then we used Backward Elimination Method. After using backward elimination method we got our Reduced model, So we can see that Life˙expectancy is not in our final regression Model. In here model was reduced under consideration of R squared values and AIC values.The lower values of AIC and BIC are generally better as they indicate a good balance between model fit. It's R squared value was 92.703 percent and AIC value is 2997.608.

When we doing hypothesis testing, in Hypothesis 01, we could understand the relationship between each independent variables and GDP per capita across countries and we could find the answers for our research questions.In Hypothesis 02 we could know our reduced model is suitable than our full model.

6.2 Conclusion

When analyzing our regression model to predict the socioeconomic determinants of GDP per capita across countries, we excluded the variable (Life expectancy) , indicating that this factor did not significantly impact GDP per capita of a country when considering the overall set of predictors. The model explains 92.7% of the variance in GDP per capita.This is a very strong fit,indicating that a large proportion of the variation in GDP per capita is explained by these predictors.

Key findings that, Income and exports showed the strongest positive association with GDP per capita, and Health expenditure had a moderate positive effect. Inverse Child mortality and Inverse total fertility rate were positively linked, means Child mortality rate and Total fertility rate have negative impact on GDP per capita.Also imports and inflation were negatively associated with the GDP per capita of a country.

Ultimately, this reduced model Indicate that the countries with higher incomes, robust export sectors, lower child mortality and lower fertility rates tend to have higher GDP per capita. Also these findings highlight the necessity of policies that increase income and exports, improve healthcare to reduce child mortality and carefully manage import dependency and inflation.

References

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APPENDIX A

Data set

You can find the final dataset for this analysis at the following link: Click here to open the dataset

Data preparation using R programming

```
data <- read.csv("Country_Data_original.csv")
library(ggplot2)
library(tidyr)
library(dplyr)

colnames(data)
head(data)
colnames(data)
summary(data)

library(psych)
library(knitr)
library(kableExtra)

# Descriptive statistics for all variables
desc_stats <- summary(data)
print(desc_stats)
# Load your dataset
data <- read.csv("Country_Data_original.csv")

# Create a custom function to extract required summary stats
custom_summary <- function(x) {
  stats <- quantile(x, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
  mean_val <- mean(x, na.rm = TRUE)
  return(c(Min = stats[1],
           `1st Qu` = stats[2],
           Median = stats[3],
```

```
            Mean = mean_val,
            `3rd Qu` = stats[4],
            Max = stats[5]))
}

# Apply the function to numeric variables only
numeric_vars <- sapply(data, is.numeric)
summary_table <- sapply(data[, numeric_vars], custom_summary)

# Transpose for better display
summary_table <- t(summary_table)

# Round for neat output
summary_table <- round(summary_table, 2)

# Print the summary table
print(summary_table)

#box plots
library(ggplot2)
library(tidyr)
library(dplyr)
colnames(data)

# Reshape data to long format
long_data <- data %>%
  select(child_mort) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to =
    "Value")

# ggplot boxplot with gray colors
ggplot(long_data, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot() +
  scale_fill_manual(values = c("child_mort" = "slategray")) +
```

```
  labs(title = "Boxplot of child mortility rates ", x = "", y = "Value"
   ) +
  theme_minimal()

#Histogram
install.packages("RColorBrewer")
library(RColorBrewer)
colnames(data)

# Create histogram of GDP per capita
ggplot(data, aes(x =gdpp )) +
  geom_histogram( binwidth= 500, fill = "steelblue", color = "black") +
  labs(title = "Histogram of GDP per capita",
       x = "GDP per capita",
       y = "Frequency") +
  theme_minimal()
## Plot histogram with density curvecolnames(data)
# Load necessary libraries


#Reggression line with scatter plot
colnames(data)
# Scatter plot with regression line
ggplot(data, aes(x = life_exp, y = gdpp)) +
  geom_point(color = "navyblue", alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", size = 1)
    +
  labs(title = "Scatter Plot of GDP per Capita vs Life expectancy",
       x = "life expectancy",
       y = "GDP per Capita") +
  theme_minimal()

#cormatrix
library(corrplot)
library(readr)
```

```r
library(dplyr)

# Select only numeric columns including gdpp and independent variables
numeric_data <- select_if(data, is.numeric)

# Optional: check variable names
names(numeric_data)
# Compute correlation matrix
cor_matrix <- cor(numeric_data, use = "complete.obs", method = "pearson")

# View the matrix
print(round(cor_matrix, 2))

# Correlation plot
corrplot(cor_matrix,
         method = "color",        # Use colored squares
         type = "upper",          # Show upper triangle only
         addCoef.col = "black",   # Add coefficient text in black
         tl.col = "black",        # Text label color
         number.cex = 0.7,        # Size of the number labels
         tl.cex = 0.8,            # Size of variable labels
         diag = FALSE)

# Build the multiple linear regression model
model <- lm(gdpp ~ income + imports +exports+ child_mort + health + inflation + life_exp + total_fer +inv_child_mort+inv_total_fer, data = data)
colnames(data)

# Calculate VIF values
vif_values <- vif(model)

# Print the VIFs
print(vif_values)
```

```
126 #
    ..........................................................................

127 ggplot(data, aes(x = income)) +
128   geom_histogram(aes(y = ..density..),
129                  binwidth = 1000,       # Adjust bin width as needed
130                  fill = "steelblue",
131                  color = "black",
132                  alpha = 0.7) +
133   geom_density(color = "red", size = 1) +
134   labs(title = "Histogram with Density Curve of Income",
135        x = "Income",
136        y = "Density") +
137   theme_minimal()
138 #
    ..........................................................................

139 ggplot(data, aes(x = exports)) +
140   geom_histogram(aes(y = ..density..),
141                  bins = 30  ,       # Adjust binwidth if needed
142                  fill = "steelblue",
143                  color = "black",
144                  alpha = 0.7) +
145   geom_density(color = "red", size = 1.15) +
146   labs(title = "Histogram of Total Fertility Rate",
147        x = "Exports (% of GDP per capita)",
148        y = "Density") +
149   theme_minimal()
150 #.....................................................
151 ggplot(data, aes(x = health)) +
152   geom_histogram(aes(y = ..density..), bins = 30, fill = "steelblue",
     color = "black") +
153   geom_density(color = "red", size = 1) +
154   labs(title = "Histogram of Health ",
155        x = "Health Spending per Capita (%)",
```

```
156         y = "Density")
```

APPENDIX B
C codes

B.1 Multiple linear regression in R programming

```
1
2  library(readr)
3  library(olsrr)
4  library(corrplot)
5
6  data <- read.csv("Country_Data_original.csv",header=TRUE,
       stringsAsFactors =FALSE)
7  data
8  colnames(data)
9
10 df<-data.frame(data)
11 df
12
13 numericc_data <- df[sapply(df, is.numeric)]
14 numericc_data
15
16 model <- lm(gdpp ~ child_mort + exports + health + imports + income +
       inflation+life_exp+total_fer+inv_child_mort+inv_total_fer, data = df
       )
17 summary(model)
18
19 library(dplyr)
20 library(knitr)
21
22 #correlation matrix
23 numeric_data <- df %>%
24     select_if(is.numeric)
25 correlation_matrix <- round(cor(numeric_data),4)
26 correlation_matrix
27
28 kable(correlation_matrix,
```

```
      caption = "Correlation Table of Numeric Variables")
summary_data <- summary(df)
summary_data

#residual plot
plot(model, which = 1)

#q-q plot for normality
qqnorm(residuals(model))
qqline(residuals(model), col = "red")

dependent_variable <- df$gdpp
hist(rstandard(model),
     main = "Histogram of Residuals",
     xlab = "Residuals",
     col = "skyblue",
     border = "white")

#Homoscedasticity
plot(fitted(model),rstandard(model),
     main = "Residuals vs Fitted (Check Homoscedasticity)",
     xlab = "Fitted values",
     ylab = "Residuals",
     pch =19, col = "navyblue")
abline(h = 0, col = "firebrick", lwd = 2)
# Get the standardized residuals from the model

library(ggplot2)

ggplot(data.frame(fitted = fitted(model), resid =rstandard(model) ),
       aes(x = fitted, y = resid)) +
  geom_point(color = "navyblue")  +
  labs(title = "Residuals vs Fitted",
       x = "Fitted values",
       y = "Standardized Residuals") +
```

```
  theme_minimal()

standardized_residuals <- rstandard(model)

# Get the fitted (predicted) values from the model
fitted_values <- fitted(model)

# Plot the standardized residuals against the fitted values
plot(fitted_values, standardized_residuals,
     main = "Standardized Residuals vs Fitted Values",
     xlab = "Fitted (Predicted) Values",
     ylab = "Standardized Residuals")

# Add a horizontal line at y=0 for reference
abline(h=0,col = "red", lty = 2)

#full model
model <- lm(gdpp ~ exports + health + imports + income + inflation+
    life_exp+ inv_child_mort+inv_total_fer, data = df)
summary(model)
anova(model)
coef(model)

library(olsrr)

#forward elimination
ols_step_forward_p(model,pent=0.05,details = T)
el.forward<- ols_step_forward_p(model,pent=0.05,details = F)
el.forward

#back elimination
ols_step_backward_p(model,perm=0.05,details = T)
el.backward<- ols_step_forward_p(model,pent=0.05,details = F)
el.backward

```

```
#best subset
ml.subset<-ols_step_best_subset(model)
ml.subset

#Full model
full_model <- lm(gdpp ~ exports + health + imports + income + inflation
   +life_exp+ inv_child_mort+inv_total_fer, data = df)
summary(full_model)$r.squared
summary(reduced_model)$r.squared

#Reduced model
reduced_model <- lm(gdpp ~ exports + health + imports + income +
   inflation + inv_child_mort+inv_total_fer, data = df)
reduced_model

summary(model_final)
anova(model_final)

anova(full_model)
anova(reduced_model)

summary(full_model)$r.squared
summary(reduced_model)$r.squared

#mean square error of full model
mse_full <- deviance(full_model) / df.residual(full_model)
mse_full

cmp<-anova(reduced_model,full_model)
cmp
df1 <- cmp$Df[2]
df1
df2 <- df.residual(full_model)
df2
alpha <- 0.05
```

```
131
132 # Critical F-value
133 f_critical <- qf(1 - alpha, df1, df2)
134 f_critical
135
136 #VIF values
137 library(car)
138 vif(full_model)
139 vif(reduced_model)
```